La Scuola d'Italia / Dr. Huson / IB Math 12 (a)
28 October 2026

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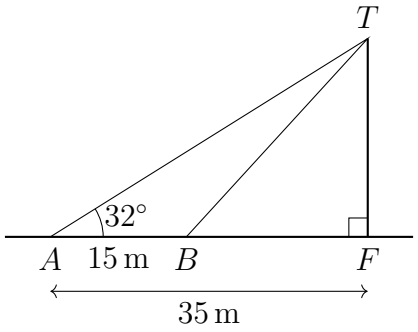
2.6 Quiz: Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad c^2 = a^2 + b^2 - 2ab\cos C \qquad \text{Area} = \frac{1}{2}ab\sin C$$

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. When a question gives an angle in radians, give angles in radians. Diagrams are not to scale.

1. [Maximum mark: 7]

   A vertical tree FT has its foot F on horizontal ground. Point A is on the ground, 35 m from F, and the angle of elevation of the top T from A is 32°. Point B lies on $[AF]$, 15 m from A.



   (a) Find the height of the tree. [2]

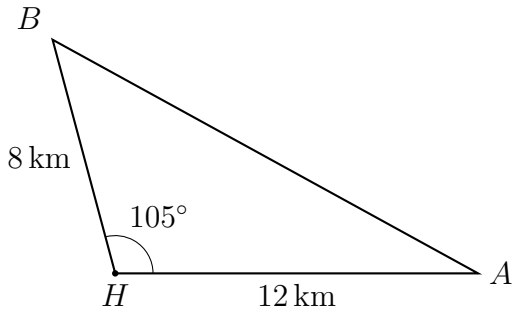
   (b) Find the distance AT. [2]

   (c) Find the angle of elevation of T from B. [3]

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2. [Maximum mark: 8]

Two boats leave a harbour H. Boat A sails 12 km in a straight line and boat B sails 8 km in a straight line. The angle between their paths is $A\hat{H}B = 105°$.



(a) Find the distance between the two boats. [3]

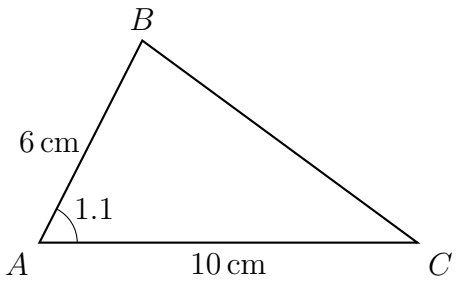
(b) Find $H\hat{A}B$. [3]

(c) Find the area of triangle HAB. [2]

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3. [Maximum mark: 8]

The following diagram shows triangle ABC, with $AB = 6\,\text{cm}$, $AC = 10\,\text{cm}$ and $B\hat{A}C = 1.1$ radians.



(a) Find BC. [3]

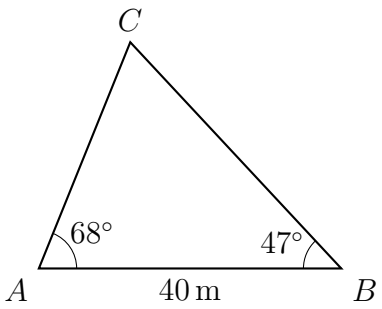
(b) Find the area of triangle ABC. [2]

(c) Find $A\hat{C}B$. [3]

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4. [Maximum mark: 9]

A triangular plot of land ABC has $AB = 40\,\text{m}$, $B\hat{A}C = 68°$ and $A\hat{B}C = 47°$.



(a) Write down $A\hat{C}B$. [1]

(b) Find AC. [3]

(c) Find the area of the plot. [2]

A second triangular plot has two sides of length 24 m and 35 m. Its area is $320\,\text{m}^2$, and the angle between these two sides is obtuse.

(d) Find the angle between these two sides. [3]